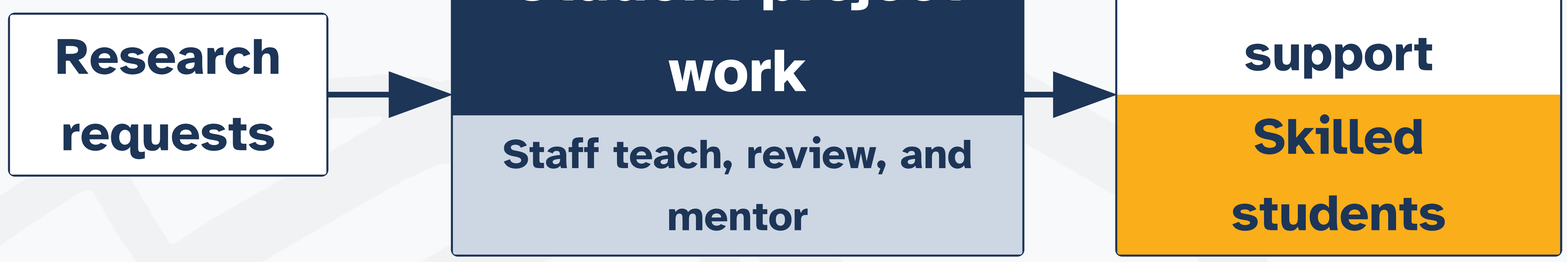
DataSquad: An entry point for RSE

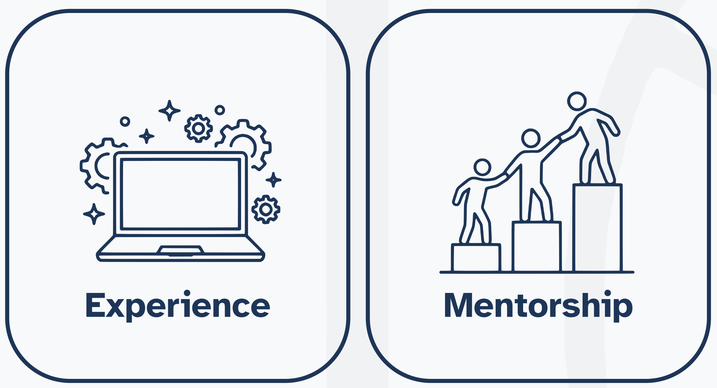
Student RSE model for research support, experiential learning, and technical workforce development

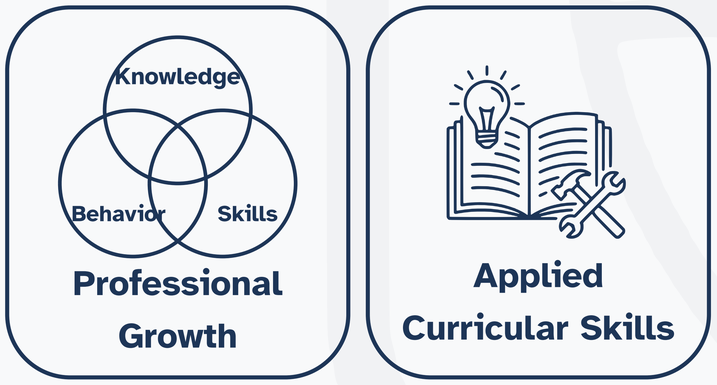
Paula Lackie, Oliver Hall, Tim Dennis, Leigh Phan
Carleton College DataSquad | UCLA Library Data Science Center DataSquad

Paid students contribute to research and build RSE skills while they do it.

Campuses need more data and software help than staff can provide. Students want this experience, but their courses can't cover everything. DataSquads fill this gap.







DataSquad roles

Project Management Intern
Triages requests, helps manage and mentor their peers, advises the staff

Data Analysts and Developers
Do the hands-on work with data and code

Technical Writers
Document projects and assure clear communications.

Staff
Supervise, mentor, and situate the DataSquad in their institution

What the work looks like

CARLETON Election-news scraper. One student built it, six improved it in 2022, and two more extended it from 2023 to 2024 for a political science project that set out to scrape over 500,000 transcripts. Their changes made long runs faster and easier to resume.

Inherited scraper		Student improvements
Colab notebook	→	Organized Python files
Long query waits	→	Query time halved
No saved progress	→	Skip completed queries

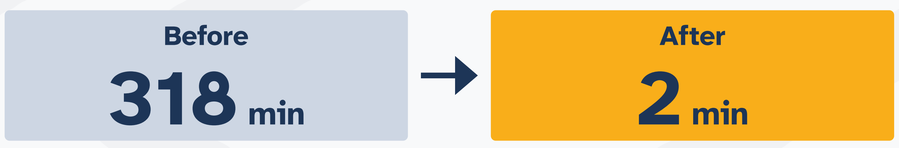
Improvements reported by the project team. Source: Carleton ITS blog, Election News Analysis project, June 2024.

UCLA Police helicopter flight data. Two students brought software skills to a research lab's R code. They refactored it with staff, and the code went through code review. Loading the data fell from about 318 minutes to about 2.



LAPD helicopter, for illustration. Photo: Chris Yarzab, CC BY 2.0, via Wikimedia Commons. Cropped and resized.

Time to load the data



Student contributions to research software. Students started wbcheck (**doi:10.5281/zenodo.23329531**) and contributed to the UC OSPO Education site (**doi:10.5281/zenodo.23330022**). Staff expanded wbcheck in 2026 with AI assistance. Depending on project needs, we add run instructions, dependency documentation, tests, and citation files.

Where students go next

Where former student employees work now, in the words of their job titles.



Words in the job titles of 96 alumni (UCLA and Carleton), from public profiles in 2026. Larger means more people; words used by fewer than 3 people are not shown. Navy is technology, steel is data and research, amber is other. Descriptive, not a placement rate.

What adopting takes

Adapt this for your environment! You'll need:

- Students and a wage budget
- To care about developing novices
- Patience

Start a DataSquad

Get the starter kit for roles, onboarding, confidentiality, and request tracking. Students learn software sharing from our open lesson, **Sharing Research Software**. We can help with supervising students: onboarding, review, attrition, expectations.

datasquad.info/join

doi:10.5281/zenodo.23381638